

Self-Attention Intuition: What Do Q, K, V Actually Compute?

Ref: [Vaswani et al., 2017]

Library Analogy

You walk into a library to find books

Q (Query) = Your question

"I want a book about attention mechanisms"

K (Key) = Book category labels

☐ Deep Learning ☐ Attention ☐ NLP ☐ Python Programming

V (Value) = Book contents

The truly valuable knowledge in each book

Matching process:

Your question(Q) $\times$ Book labels(K) $\rightarrow$ Match scores

High-scoring books' content(V) $\rightarrow$ Information you get

Computation Complexity Comparison

Self-Attention: $O(L^2 \times d)$

Every pair of tokens computes attention

Long sequences cause computation explosion $\rightarrow$ Infeasible at $L=100K$

RNN/LSTM: $O(L \times d^2)$

Linear in sequence length

But cannot parallelize, poor long-range dependencies

Mathematical Formulation

Input: $X \in \mathbb{R}^{(L \times d)}$

$Q = X \cdot W_Q$, $K = X \cdot W_K$, $V = X \cdot W_V$

$W_Q, W_K, W_V \in \mathbb{R}^{(d \times d_k)}$ Learnable parameters

Attention scores (scaled dot-product):

$S = (Q \cdot K^T) / \sqrt{d_k}$

Divide by $\sqrt{d_k}$ to prevent dot products from growing too large and saturating softmax
[Vaswani et al., 2017] original paper Section 3.2.1

Add causal mask (Decoder):

$S = S + M$, $M_{ij} = -\infty$ if $i < j$

Prevents position i from seeing position j ($j > i$)

Final output:

Scaled Dot-Product Attention Formula

$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) \cdot V$

Q, K, V are the query, key, value matrices

d_k is the Key vector dimension (scaling factor)

softmax ensures weights are non-negative and sum to 1

[Vaswani et al., 2017] Eq. (1)